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United States Patent	12415078
Kind Code	B2
Date of Patent	September 16, 2025
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Method and apparatus for charge balancing during delivery of electrical stimulation

Abstract

A medical device is configured to deliver a series of electrical stimulation pulses including opposing polarity pulses. The medical device delivers a charge balancing pulse by modifying every nth pulse of the electrical stimulation pulses to reduce a net charge delivered over the series of electrical stimulation pulses. In some examples, the medical device may be an implantable medical device that is coupled to an extra-cardiovascular lead for delivering the cardiac pacing pulses.

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Appl. No.: 17/840341

Filed: June 14, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220314003 A1	Oct. 06, 2022

Related U.S. Application Data

continuation parent-doc US 15964355 20180427 US 11369795 child-doc US 17840341

Publication Classification

Int. Cl.: A61N1/36 (20060101); A61N1/362 (20060101); A61N1/37 (20060101)

U.S. Cl.:

CPC **A61N1/36167** (20130101); **A61N1/36125** (20130101); **A61N1/36153** (20130101);
A61N1/3624 (20130101); A61N1/36139 (20130101); A61N1/3627 (20130101);
A61N1/3706 (20130101)

Field of Classification Search

CPC: A61N (1/36167); A61N (1/36125); A61N (1/36153); A61N (1/3624); A61N (1/36139);
A61N (1/3627); A61N (1/3706)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5083562	12/1991	de Coriolis et al.	N/A	N/A
5601608	12/1996	Mouchawar	N/A	N/A
5718718	12/1997	Kroll et al.	N/A	N/A
6671546	12/2002	Cansell et al.	N/A	N/A
8195291	12/2011	Norton et al.	N/A	N/A
8340762	12/2011	Vonk et al.	N/A	N/A
9278229	12/2015	Reinke et al.	N/A	N/A
9421385	12/2015	Vijayagopal et al.	N/A	N/A
9656076	12/2016	Trier et al.	N/A	N/A
9855414	12/2017	Marshall et al.	N/A	N/A
9937344	12/2017	Starkebaum et al.	N/A	N/A
11369795	12/2021	Corndorf	N/A	A61N 1/3624
2008/0281375	12/2007	Chen	607/40	A61N 1/36007
2009/0270943	12/2008	Maschino	N/A	N/A
2012/0010679	12/2011	Edvinsson	N/A	N/A
2012/0197356	12/2011	Wei	607/74	A61N 1/36171
2013/0282079	12/2012	Kallmyer	607/62	A61N 1/36125
2014/0005753	12/2013	Carbunaru	N/A	N/A
2014/0107720	12/2013	Bornzin	607/9	A61N 1/3627
2014/0277267	12/2013	Vansickle et al.	N/A	N/A
2015/0306375	12/2014	Marshall et al.	N/A	N/A
2015/0360041	12/2014	Stahmann et al.	N/A	N/A
2016/0158567	12/2015	Marshall et al.	N/A	N/A
2016/0228708	12/2015	Ternes et al.	N/A	N/A
2017/0157399	12/2016	Anderson et al.	N/A	N/A
2017/0157413	12/2016	Anderson et al.	N/A	N/A
2017/0259065	12/2016	Baru et al.	N/A	N/A
2018/0221677	12/2017	Grinberg	N/A	A61N 1/3987
2018/0272124	12/2017	Kibler	N/A	A61N 1/025
2019/0001139	12/2018	Mishra	N/A	A61N 1/37229

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017096143	12/2016	WO	N/A

OTHER PUBLICATIONS

(PCT/US2019/029086) PCT Notification of Transmittal of the International Search Report and the Written Opinion of the International Searching Authority, Mailed Sep. 9, 2019, 12 pages. cited by applicant

Wang, Liquan, Pacemaker pacing function with ECG, Journal of Clinical Electrophysiology, Dec. 15, 2014, pp. 6-15, No. 6. cited by applicant

First Office Action for CN Application No. 201980028474.8 dated Jan. 25, 2024, 12 pages. cited by applicant

Primary Examiner: Hulbert; Amanda K

Background/Summary

REFERENCE TO RELATED APPLICATION (1) This application is a Continuation of U.S. patent application Ser. No. 15/964,355, filed on Apr. 27, 2018, published as U.S. Publication No. 2019/0329041, and incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to a medical device and method for charge balancing electrical stimulation pulses delivered to a patient, such as cardiac pacing pulses.

BACKGROUND

(2) A variety of medical devices exist for delivering an electrical stimulation therapy, monitoring a physiological condition of a patient or a combination thereof. Some of these medical devices include implantable medical devices (IMDs) that deliver electrical stimulation pulses using implanted electrodes. In some examples, IMDs may employ one or more elongated electrical leads carrying stimulation electrodes, sensing electrodes, and/or other sensors. IMDs may deliver therapy to or monitor conditions of a variety of organs, nerves, muscle or tissue, such as the heart, brain, stomach, spinal cord, pelvic floor, or the like. Implantable medical electrical leads may be configured to position electrodes or other sensors at desired locations for delivery of electrical stimulation or sensing of physiological conditions. For example, electrodes or sensors may be carried along a distal portion of a lead that is extended subcutaneously, submuscularly, intra-thoracically, intra-abdominally, intra-cranially, or transvenously. A proximal portion of the lead may be coupled to the housing of the IMD, providing electrical connection to circuitry contained within the housing such as signal generation circuitry and/or sensing circuitry.

(3) Some IMDs, such as cardiac pacemakers or implantable cardioverter defibrillators (ICDs), provide therapeutic electrical stimulation to the heart of the patient via electrodes carried by one or more implantable leads and/or the housing of the pacemaker or ICD. The leads may be transvenous, e.g., advanced into the heart through one or more veins to position endocardial electrodes in intimate contact with the heart tissue. Other leads may be non-transvenous leads implanted outside the heart, e.g., implanted epicardially, pericardially, or subcutaneously. The electrodes are used to deliver electrical pulses to the heart to address abnormal cardiac rhythms.

(4) IMDs capable of delivering electrical pulses for treating abnormal cardiac rhythms typically sense signals representative of intrinsic depolarizations of the heart and analyze the sensed signals to identify the abnormal rhythms. Upon detection of an abnormal rhythm, the device may deliver an appropriate electrical stimulation therapy to restore a more normal rhythm. For example, a pacemaker or ICD may deliver pacing pulses to the heart upon detecting bradycardia or tachycardia. An ICD may deliver high voltage cardioversion or defibrillation shocks to the heart upon detecting fast ventricular tachycardia or fibrillation.

SUMMARY

(5) The techniques of this disclosure generally relate to a medical device and method for charge balancing electrical stimulation pulses delivered to a patient's body tissue, such as cardiac pacing pulses or neurostimulation pulses, via one or more electrodes. Electrode corrosion can occur when electrical stimulation pulses delivered by the electrodes deliver a net electrical charge that accumulates over time. This effect can be minimized by charge balancing the electrical stimulation pulses to thereby reduce or eliminate the net charge that accumulates due to the delivered pulses. A medical device operating according to the techniques disclosed herein controls the polarity, amplitude and pulse width of each electrical stimulation pulse in order to provide charge balancing over a series of electrical stimulation pulses.

(6) In one example, the disclosure provides a medical device having a therapy delivery circuit and a control circuit. The therapy delivery circuit is configured to generate and deliver electrical stimulation pulses via electrodes coupled to the therapy delivery circuit. The control circuit is coupled to the therapy delivery circuit and is configured to control the therapy delivery circuit to deliver electrical stimulation pulses comprising opposing polarity electrical stimulation pulses and deliver a charge balancing pulse by modifying every nth pulse of the electrical stimulation pulses. The charge balancing pulse is delivered to reduce a net charge delivered over the electrical stimulation pulses.

(7) In another example, the disclosure provides a method performed by a medical device. The method includes delivering electrical stimulation pulses comprising opposing polarity electrical stimulation pulses and delivering a charge balancing pulse by modifying every nth pulse of the electrical stimulation pulses. The charge balancing pulse is delivered to reduce a net charge delivered over the electrical stimulation pulses.

(8) In yet another example, the disclosure provides a non-transitory, computer-readable medium storing a set of instructions which when executed by a control circuit of a medical device, cause the medical device to deliver electrical stimulation pulses comprising opposing polarity electrical stimulation pulses and deliver a charge balancing pulse by modifying every nth pulse of the electrical stimulation pulses. The charge balancing pulse is delivered to reduce a net charge delivered over the electrical stimulation pulses.

(9) The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIGS. 1A and 1B are conceptual diagrams of an extra-cardiovascular ICD system according to one example.

(2) FIG. 2A is a front view and FIG. 2B is a side view of a patient implanted with an ICD system according to another example.

(3) FIG. 3 is a schematic diagram of an ICD according to one example.

(4) FIG. 4 is a schematic diagram of high voltage therapy circuit coupled to a processor and HV therapy control circuit.

(5) FIG. 5 is a conceptual diagram of a low voltage therapy circuit according to one example.

(6) FIG. 6 is a conceptual diagram of a series of alternating-polarity pacing pulses including a charge balancing pulse.

(7) FIG. 7 is a diagram of cardiac pacing pulses delivered according to another example technique for charge balancing.

(8) FIG. 8 is a diagram of another technique for delivering charge balancing cardiac pacing pulses.

(9) FIG. 9 is a flow chart of a method for delivering charge balancing cardiac pacing pulses according to one example.

(10) FIG. 10 is a conceptual diagram of one example of a lookup table that may be stored in the memory of an ICD.

(11) FIG. 11 is a flow chart of a method for establishing a modification of a charge balancing pacing pulse by an ICD according to one example.

DETAILED DESCRIPTION

(12) In general, this disclosure describes techniques for charge balancing electrical stimulation pulses delivered to a patient. For example, the techniques may be particularly useful in the delivery of cardiac pacing pulses but may be applied in the delivery of neurostimulation pulses or any other electrical stimulation pulses delivered to a patient's body tissue via medical electrodes. The delivered electrical stimulation pulses may be delivered for therapeutic purposes, such as cardiac pacing pulses, but may be delivered for diagnostic or testing purposes in some examples. The disclosed charge balancing techniques may be implemented in any medical device configured to deliver electrical stimulation pulses, particularly electrical stimulation pulses that are delivered repetitively over time resulting in an accumulated electrical charge at the stimulating electrode(s). In some examples, the charge balancing techniques are used during the delivery of repetitive, opposing polarity electrical stimulation pulses having an inherent imbalance between the charge delivered by a positive polarity pulse compared to the charge delivered by a negative polarity pulse.

(13) Electrical stimulation pulses that are charge balanced are intended to deliver a net-zero or near net-zero charge over time. Electrical stimulation pulses that are charge imbalanced, meaning that a non-net zero charge accumulates at the tissue-electrode interface over time, can lead to electrode corrosion and cause polarization artifact that can interfere with sensing of cardiac electrical signals. Charge accumulation may occur when the electrical stimulation pulses are imbalanced during delivery of relatively high energy pulses (e.g., high current or voltage amplitude and/or long pulse duration) as well as relatively low energy pulses (e.g., low current or voltage amplitude and/or short pulse duration).

(14) Relatively high pulse energy pulses are delivered for cardiac pacing when a cardiac capture threshold is high. The capture threshold is the minimum pulse energy that captures the heart, causing a non-intrinsic depolarization or evoked response. Factors that may contribute to a high cardiac capture threshold include pacing lead impedance and electrode location. For example, when the heart is being paced using electrodes carried by a non-transvenous, extra-cardiovascular lead, the cardiac capture threshold is significantly higher than the cardiac capture threshold when the heart is being paced by endocardial electrodes carried by a transvenous lead, which are in intimate contact with myocardial tissue. As used herein, the term “extra-cardiovascular” refers to a position outside the blood vessels, heart, and pericardium surrounding the heart of a patient. Implantable electrodes carried by extra-cardiovascular leads may be positioned extra-thoracically (outside the ribcage and sternum) or intra-thoracically (beneath the ribcage or sternum) but may not be in intimate contact with myocardial tissue. The charge balancing techniques disclosed herein, therefore, may be used in some examples for delivering extra-cardiovascular pacing pulses or any pacing pulses delivered using electrodes that are not in direct contact with the heart.

(15) Even relatively low energy electrical stimulation pulses, e.g., cardiac pacing pulses delivered using endocardial or epicardial electrodes in direct contact with the heart, if imbalanced, can lead to electrode corrosion over time. Charge balancing the pacing pulses may minimize these effects. In endocardial or epicardial pacing applications, the effects of imbalanced charge delivery may be mitigated by delivering a non-therapeutic, recharge pulse having the opposite polarity of the pacing pulse during the physiological refractory period of the myocardium that follows the pacing pulse. The opposite polarity recharge pulse may not be precisely charge-equivalent to the pacing pulse but may adequately prevent electrode corrosion since in endocardial or epicardial pacing, charge accumulation over time is small because the charge delivered during each pulse is relatively small.

In extra-cardiovascular cardiac pacing systems, the pacing pulse energy delivered to capture the heart may be on the order of at least twice and even 10 to 100 times greater per pacing pulse than during endocardial pacing. For example, an endocardial pacing pulse may be up to 8.5 Volts in amplitude and up to 1.5 ms in pulse width and is more typically less than 5 Volts, e.g., 1 V, in amplitude and 0.5 ms in pulse width. An extra-cardiovascular pacing pulse, on the other hand, may be 10 to 40 Volts in amplitude and up to 8 ms in pulse width. As such, without charge balancing, the corrosive process acting on the pacing electrodes may be amplified 100 times or more during extra-cardiovascular pacing than during endocardial pacing. Because the non-therapeutic recharge pulses that may be applied in endocardial pacing applications are generally not precisely charge-equivalent to the therapeutic pacing pulses, the technique of using recharge pulses may not provide adequate charge balancing in all applications, particularly during delivery of high pulse energy electrical stimulation pulses because of the larger charge accumulation and corrosive effect over time.

(16) One approach to providing charge balancing in high voltage, extra-cardiovascular pacing applications is to deliver therapeutic pulses having alternating polarity in order to provide charge balancing over each pair of pacing cycles. Examples of charge balanced pacing pulses delivered by a high voltage pacing circuit are generally disclosed in U.S. patent application Ser. No. 15/425,169 (Grinberg et al., filed Feb. 6, 2017), incorporated herein by reference in its entirety. However, in some systems, inherent imbalances of opposite polarity pulses may exist such that alternating therapeutic pulses may still result in a net charge accumulation over time and associated risk of electrode corrosion, especially in the high voltage pacing applications such as extra-cardiovascular pacing. For these reasons, charge balancing during relatively high voltage (or high current amplitude) pacing applications for preserving the integrity of the pacing system presents a greater challenge than during endocardial pacing applications which use much lower pacing pulse voltage (or current) amplitudes. Charge balancing techniques are disclosed herein for providing cardiac pacing by an IMD while preventing or minimizing electrode corrosion over time. These techniques may include delivering a series of cardiac pacing pulses having opposing polarities, which are alternating positive and negative polarity pulses in some examples, with every nth pacing pulse in the series being modified to balance a net charge delivered during the preceding one or more pulses in the series.

(17) FIGS. 1A and 1B are conceptual diagrams of an extra-cardiovascular ICD system **10** according to one example. FIG. 1A is a front view of ICD system **10** implanted within patient **12**. FIG. 1B is a transverse sectional view of ICD system **10** implanted within patient **12**. ICD system **10** includes an ICD **14** connected to an extra-cardiovascular electrical stimulation and sensing lead **16**. An “extra-cardiovascular lead” as used herein, refers to a lead that is implanted outside the heart and blood vessels of the patient's cardiovascular system. An extra-cardiovascular lead may extend subcutaneously, submuscularly or intra-thoracically, for example. FIGS. 1A and 1B are described in the context of an ICD system **10** capable of providing cardioversion/defibrillation (CV/DF) shocks and pacing pulses. It is contemplated that the techniques disclosed herein for providing charge balancing of cardiac pacing pulses, however, may be implemented in systems that are not configured for delivering shock therapies. The disclosed charge balancing cardiac pacing techniques may be implemented in any medical device configured to deliver cardiac pacing pulses and may also be beneficial in any medical device that delivers electrical stimulation pulses, particularly when alternating polarity pulses, or more generally a series of pulses comprising opposing polarity pulses, are delivered having an inherent imbalance in the charge delivered by the positive and negative pulses.

(18) ICD **14** includes a housing **15** that forms a hermetic seal that protects internal components of ICD. Internal components includes pulse generating circuitry, which may include a high voltage therapy circuit, for generating high voltage CV/DF shocks and high voltage pacing pulses, and a low voltage therapy circuit, for generating relatively lower voltage pacing pulses delivered using

extra-cardiovascular electrodes carried by lead **16**. The housing **15** of ICD **14** may be formed of a conductive material, such as titanium or titanium alloy. The housing **15** may function as a housing electrode (sometimes referred to as a can electrode). In examples described herein, housing **15** may be used as an active can electrode for use in delivering CV/DF shocks or other high voltage pulses, including cardiac pacing pulses, delivered using the high voltage therapy circuit. In other examples, housing **15** may be available for use in delivering unipolar, cardiac pacing pulses delivered using the low voltage therapy circuit in conjunction with a lead-based cathode electrode. In other instances, the housing **15** of ICD **14** may include a plurality of electrodes on an outer portion of the housing. The outer portion(s) of the housing **15** functioning as an electrode(s) may be coated with a material, such as titanium nitride to reduce post-stimulation polarization artifact.

(19) ICD **14** includes a connector assembly **17** (also referred to as a connector block or header) that includes electrical feedthroughs crossing housing **15** to provide electrical connections between conductors extending within the lead body **18** of lead **16** and electronic components included within the housing **15** of ICD **14**. As will be described in further detail herein, housing **15** may house one or more processors, memories, transceivers, sensors, cardiac electrical signal sensing circuitry, therapy delivery circuitry, power sources and other components for sensing cardiac electrical signals, detecting a heart rhythm, and controlling and delivering electrical stimulation pulses to treat an abnormal heart rhythm.

(20) Lead **16** includes an elongated lead body **18** having a proximal end **27** that includes a lead connector (not shown) configured to be connected to ICD connector assembly **17** and a distal portion **25** that includes one or more electrodes. In the example illustrated in FIGS. **1A** and **1B**, the distal portion **25** of lead **16** includes defibrillation electrodes **24** and **26** and pace/sense electrodes **28** and **30**. In some cases, defibrillation electrodes **24** and **26** may together form a defibrillation electrode in that they may be configured to be activated concurrently. Alternatively, defibrillation electrodes **24** and **26** may form separate defibrillation electrodes in which case each of the electrodes **24** and **26** may be activated independently.

(21) Electrodes **24** and **26** (and in some examples housing **15**) are referred to herein as defibrillation electrodes because they are utilized, individually or collectively, for delivering high voltage stimulation therapy (e.g., cardioversion or defibrillation shocks). Electrodes **24** and **26** may be elongated coil electrodes and generally have a relatively high surface area for delivering high voltage shock therapy pulses compared to the surface area of the pacing and sensing electrodes **28** and **30**, which may be used to deliver cardiac pacing pulses that are lower in pulse energy than CV/DF shock pulses. However, electrodes **24** and **26** and housing **15** may also be utilized to provide pacing functionality, sensing functionality or both pacing and sensing functionality in addition to or instead of high voltage CV/DF shock therapy. In this sense, the use of the term “defibrillation electrode” herein should not be considered as limiting the electrodes **24** and **26** for use in only CV/DF shock therapy applications. Electrodes **24** and **26** may be used in a pacing electrode vector for delivering extra-cardiovascular pacing pulses such as but not limited to bradycardia pacing pulses, anti-tachycardia pacing (ATP) pulses, entrainment pulses for tachyarrhythmia induction, cardiac resynchronization therapy (CRT) pulses, and/or post-shock pacing pulses. Electrodes **24** and **26** may also be used in a sensing vector used to sense cardiac electrical signals and detect abnormal heart rhythms, such as ventricular tachycardia (VT) and ventricular fibrillation (VF).

(22) Electrodes **28** and **30** are relatively smaller surface area electrodes for delivering relatively lower voltage pacing pulses and for sensing cardiac electrical signals. Electrodes **28** and **30** are referred to herein as pace/sense electrodes because they are generally configured for use in relatively lower voltage applications than defibrillation electrodes **24** and **26**. Electrodes **28** and **30** may be used as either a cathode or anode for delivery of pacing pulses, including any of the examples given above, and/or sensing of cardiac electrical signals. In some instances, electrodes **28** and **30** may provide only pacing functionality, only sensing functionality or both. While electrodes

28 and **30** are generally referred to as being used for delivering relatively lower voltage pacing pulses compared to high voltage CV/DF shock pulses delivered by defibrillation electrodes **24** and **26**, it is to be understood that the cardiac pacing pulses delivered using any combination of the extra-cardiovascular electrodes **24**, **26**, **28**, **30** and housing **15** are generally higher in voltage amplitude and/or pulse width than cardiac pacing pulses delivered using endocardial or epicardial electrodes. As such, the extra-cardiovascular pacing pulses delivered by ICD system **10** are considered relatively high voltage pacing pulses (compared to pacing pulses delivered by transvenous ICD systems) and may be controlled using the charge balancing techniques disclosed herein.

(23) In the example illustrated in FIG. **1A**, electrode **28** is located proximal to defibrillation electrode **24**, and electrode **30** is located between defibrillation electrodes **24** and **26**. In other examples, electrodes **28** and **30** may be positioned at other locations along lead **16**, which may include one or more pace/sense electrodes. Electrodes **28** and **30** are illustrated as ring electrodes, however electrodes **28** and **30** may comprise any of a number of different types of electrodes, including ring electrodes, short coil electrodes, hemispherical electrodes, directional electrodes, or the like.

(24) Lead **16** extends subcutaneously or submuscularly over the ribcage **32** medially from the connector assembly **17** of ICD **14** toward a center of the torso of patient **12**, e.g., toward xiphoid process **20** of patient **12**. At a location near xiphoid process **20**, lead **16** bends or turns and extends superiorly within anterior mediastinum **36** in a substernal position. Anterior mediastinum **36** (seen in FIG. **1B**) may be viewed as being bounded laterally by pleurae **39**, posteriorly by pericardium **38**, and anteriorly by sternum **22**. In some instances, the anterior wall of anterior mediastinum **36** may also be formed by the transversus thoracis muscle and one or more costal cartilages. Anterior mediastinum **36** includes a quantity of loose connective tissue (such as areolar tissue), adipose tissue, some lymph vessels, lymph glands, substernal musculature, small side branches of the internal thoracic artery or vein, and the thymus gland. In one example, the distal portion **25** of lead **16** extends along the posterior side of sternum **22** substantially within the loose connective tissue and/or substernal musculature of anterior mediastinum **36**. A lead implanted such that the distal portion **25** is substantially within anterior mediastinum **36** may be referred to as a “substernal lead.”

(25) In the example illustrated in FIGS. **1A** and **1B**, lead **16** is located substantially centered under sternum **22**. In other instances, however, lead **16** may be implanted such that it is offset laterally from the center of sternum **22**. In some instances, lead **16** may extend laterally such that distal portion **25** of lead **16** is underneath/below the ribcage **32** in addition to or instead of sternum **22**. In other examples, the distal portion **25** of lead **16** may be implanted in other extra-cardiovascular, intra-thoracic locations, including the pleural cavity or around the perimeter of and adjacent to the pericardium **38** of heart **8**. Other implant locations and lead and electrode arrangements that may be used in conjunction with the cardiac pacing techniques described herein are generally disclosed in U.S. patent application Publication No. 2015/0306375 (Marshall, et al.) and U.S. Pat. No. 9,855,414 (Marshall, et al.), both of which are incorporated herein by reference in their entirety. For example, as shown in FIGS. **2A** and **2B**, lead **16** may extend superiorly and subcutaneously or submuscularly over the ribcage and/or sternum **22**, rather than substernally. Alternatively, lead **16** may be placed along other subcutaneous or submuscular paths. The path of lead **16** may depend on the location of ICD **14**, the arrangement and position of electrodes carried by the lead distal portion **25**, and/or other factors.

(26) Lead body **18** may be a flexible lead body without preformed curves or bends. In other examples an extra-cardiovascular lead including one or more defibrillation electrodes and/or one or more pacing and sensing electrodes may be carried by a lead body having one or more pre-formed curves or bends or a serpentine, undulating or zig-zagging distal portion of the lead body **18**. Examples of other extra-cardiovascular leads that may be implemented with the techniques described herein are generally disclosed in U.S. patent application Publication No. 2016/0158567

(Marshall, et al.), incorporated herein by reference in its entirety. The techniques disclosed herein are not limited to any particular electrode arrangement or lead body design, however.

(27) Electrical conductors (not illustrated) extend through one or more lumens of the elongated lead body **18** of lead **16** from the lead proximal end **27** to electrodes **24**, **26**, **28**, and **30** located along the distal portion **25** of the lead body **18**. The conductors electrically couple the electrodes **24**, **26**, **28**, and **30** to circuitry, such as a therapy delivery circuit and/or a sensing circuit, of ICD **14** via connections in the connector assembly **17**, including associated electrical feedthroughs crossing housing **15**. The electrical conductors transmit therapy from a therapy circuit within ICD **14** to one or more of defibrillation electrodes **24** and **26** and/or pace/sense electrodes **28** and **30** and transmit cardiac electrical signals produced by the patient's heart **8** from one or more of defibrillation electrodes **24** and **26** and/or pace/sense electrodes **28** and **30** to the sensing circuit within ICD **14**.

(28) ICD **14** may obtain cardiac electrical signals corresponding to electrical activity of heart **8** via a combination of sensing vectors that include combinations of electrodes **28** and **30**. In some examples, housing **15** of ICD **14** is used in combination with one or more of electrodes **28** and/or **30** in a sensing electrode vector. ICD **14** may even obtain cardiac electrical signals using a sensing vector that includes one or both defibrillation electrodes **24** and/or **26**, e.g., between electrodes **24** and **26** or between one of electrodes **24** or **26** in combination with one of electrodes **28**, **30** and/or housing **15**.

(29) ICD **14** analyzes the cardiac electrical signals received from one or more of the sensing vectors to monitor for abnormal rhythms, such as bradycardia, VT and VF. ICD **14** may analyze the heart rate and/or morphology of the cardiac electrical signals to monitor for tachyarrhythmia in accordance with any of a number of tachyarrhythmia detection techniques. ICD **14** generates and delivers electrical stimulation therapy in response to detecting a tachyarrhythmia (e.g., VT or VF). ICD **14** may deliver ATP in response to VT detection, and in some cases may deliver ATP prior to a CV/DF shock or during high voltage capacitor charging in an attempt to avert the need for delivering a CV/DF shock. ATP may be delivered using an extra-cardiovascular pacing electrode vector selected from any of electrodes **24**, **26**, **28**, **30** and/or housing **15**.

(30) If ATP is not delivered or does not successfully terminate VT or when VF is detected, ICD **14** may deliver one or more CV/DF shocks via one or both of defibrillation electrodes **24** and **26** and/or housing **15**. ICD **14** may deliver the CV/DF shocks using electrodes **24** and **26** individually or together as a cathode (or anode) and with the housing **15** as an anode (or cathode).

(31) Bradycardia pacing pulses may be delivered in response to too slow of a heart rate or asystole. ICD **14** may generate and deliver cardiac pacing pulses, such as post-shock pacing pulses or bradycardia pacing pulses using a pacing electrode vector that includes one or more of the electrodes **24**, **26**, **28**, **30** and the housing **15** of ICD **14**. The timing of pacing pulses may be controlled by a controller included in ICD **14** by setting a pacing escape interval timer or counter according to a lower pacing rate or back-up pacing interval. Bradycardia pacing, post-shock pacing, CRT, atrial-tracking ventricular pacing pulses, or any other type of pacing therapy delivered by ICD **14** may be controlled using the charge balancing techniques disclosed herein.

(32) An external device **40** is shown in telemetric communication with ICD **14** by a communication link **42**. External device **40** may include a processor, display, user interface, telemetry unit and other components for communicating with ICD **14** for transmitting and receiving data via communication link **42**. Communication link **42** may be established between ICD **14** and external device **40** using a radio frequency (RF) link such as BLUETOOTH®, Wi-Fi, or Medical Implant Communication Service (MICS) or other RF or selected communication frequency bandwidth.

(33) External device **40** may be embodied as a programmer used in a hospital, clinic or physician's office to retrieve data from ICD **14** and to program operating parameters and algorithms in ICD **14** for controlling ICD functions. External device **40** may be used to program cardiac rhythm detection parameters and therapy control parameters used by ICD **14**. Control parameters used to generate and deliver cardiac pacing pulses according to the charge balancing techniques disclosed herein

may be programmed into ICD **14** using external device **40**.

(34) Data stored or acquired by ICD **14**, including physiological signals or associated data derived therefrom, results of device diagnostics, and histories of detected rhythm episodes and delivered therapies, may be retrieved from ICD **14** by external device **40** following an interrogation command. External device **40** may alternatively be embodied as a home monitor or hand held device.

(35) FIG. **2A** is a front view and FIG. **2B** is a side view of a patient implanted with ICD system **10** with extra-cardiovascular lead **16** implanted in a different location than FIGS. **1A-1B**. In some examples, extra-cardiovascular electrical stimulation therapies may be delivered using electrodes that remain outside the thoracic cavity. In the example of FIG. **2A**, the distal portion **25** of lead **16** extends over ribcage **32**, laterally offset from and substantially parallel to sternum **22**. In other examples, lead **16** may extend over sternum **22** or may extend diagonally over a portion of sternum **22** and ribcage **32**.

(36) FIGS. **1A-2B** are illustrative in nature and should not be considered limiting of the practice of the techniques disclosed herein. For instance, ICD **14** is shown implanted subcutaneously on the left side of patient **12** along the ribcage **32**. ICD **14** may, in other instances, be implanted between the left posterior axillary line and the left anterior axillary line of patient **12**. ICD **14** may be implanted at other subcutaneous or submuscular locations in patient **12** such as in a subcutaneous pocket in the pectoral region. In this case, lead **16** may extend subcutaneously or submuscularly from ICD **14** toward the manubrium of sternum **22** and bend or turn and extend inferior from the manubrium to the desired location subcutaneously or submuscularly. In yet another example, ICD **14** may be placed abdominally. Lead **16** may be implanted in other extra-cardiovascular locations as well and include other electrode and lead body configurations.

(37) FIG. **3** is a schematic diagram of ICD **14** according to one example. The electronic circuitry enclosed within housing **15** (shown schematically as a can electrode in FIG. **3**) includes software, firmware and hardware that cooperatively monitor one or more cardiac electrical signals, determine when a pacing therapy is necessary, and deliver prescribed pacing therapies as needed. The software, firmware and hardware are also configured to determine when a CV/DF shock is necessary, and deliver prescribed CV/DF shock therapies. ICD **14** is coupled to an extra-cardiovascular lead, such as lead **16** carrying extra-cardiovascular electrodes **24**, **26**, **28** and **30** for delivering pacing therapies, CV/DF shock therapies and sensing cardiac electrical signals.

(38) ICD **14** includes a control circuit **80**, memory **82**, therapy delivery circuit **84**, cardiac electrical signal sensing circuit **86**, and telemetry circuit **88**. ICD **14** may include an impedance measurement circuit **90** for delivering a drive signal across a pacing electrode vector and measuring a resulting voltage for determining an electrical impedance of the pacing electrode vector.

(39) A power source **98** provides power to the circuitry of ICD **14**, including each of the circuits **80**, **82**, **84**, **86**, **88**, **90** as needed. Power source **98** may include one or more energy storage devices, such as one or more rechargeable or non-rechargeable batteries. The connections between power source **98** and each of the other circuits **80**, **82**, **84**, **86** and **88** are to be understood from the general block diagram of FIG. **3**, but are not shown for the sake of clarity. For example, power source **98** is coupled to low voltage (LV) and high voltage (HV) charging circuits included in therapy delivery circuit **84** for charging LV and HV capacitors, respectively, or other energy storage devices included in therapy delivery circuit **84** for producing electrical stimulation pulses.

(40) The functional blocks shown in FIG. **3** represent example functionality included in ICD **14**. ICD **14** may include more or fewer components than illustrated in FIG. **3**. ICD **14** may include any discrete and/or integrated electronic circuit components that implement analog and/or digital circuits capable of producing the functions attributed to ICD **14** herein, which may include one or more of an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that execute one or more software or firmware programs, a combinational logic circuit, state machine, or other suitable components that provide the described

functionality. The particular form of software, hardware and/or firmware employed to implement the functionality disclosed herein will be determined primarily by the particular system architecture employed in the device and by the particular detection and therapy delivery methodologies employed by the ICD **14**. Providing software, hardware, and/or firmware to accomplish the described functionality in the context of any modern IMD system, given the disclosure herein, is within the abilities of one of skill in the art.

(41) Memory **82** may include any volatile, non-volatile, magnetic, or electrical non-transitory computer readable storage media, such as a random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other memory device. Furthermore, memory **82** may include non-transitory computer readable media storing instructions that, when executed by one or more processing circuits, cause control circuit **80** or other ICD circuits to perform various functions attributed to ICD **14** or those ICD circuits. The non-transitory computer-readable media storing the instructions may include any of the media listed above.

(42) Functionality attributed to ICD **14** may be performed by separate hardware, firmware or software components, or integrated within common hardware, firmware or software components. For example, cardiac pacing operations may be performed by therapy delivery circuit **84** under the control of control circuit **80** and may include operations implemented in a processor executing instructions stored in memory **82**.

(43) Control circuit **80** communicates with therapy delivery circuit **84** and sensing circuit **86** for sensing cardiac electrical activity, detecting cardiac rhythms, and controlling delivery of cardiac electrical stimulation therapies in response to sensed cardiac signals. Therapy delivery circuit **84** and electrical sensing circuit **86** are each electrically coupled to any or all of electrodes **24**, **26**, **28** and **30** carried by lead **16** shown in FIGS. **1A-2B** and the housing **15**, which may function as a common or ground electrode or as an active can electrode for delivering CV/DF shock pulses.

(44) Cardiac electrical signal sensing circuit **86** may be selectively coupled to electrodes **24**, **26**, **28** and **30** and housing **15** in order to monitor electrical activity of the patient's heart. Sensing circuit **86** may include switching circuitry for selecting which of electrodes **24**, **26**, **28** and **30** and housing **15** are coupled to sense amplifiers or other cardiac event detection circuitry included in sensing circuit **86**. Switching circuitry may include a switch array, switch matrix, multiplexer, or any other type of switching device suitable to selectively couple sense amplifiers to selected electrodes. The cardiac event detection circuitry within electrical sensing circuit **86** may include one or more sense amplifiers, filters, rectifiers, threshold detectors, comparators, analog-to-digital converters (ADCs), or other analog or digital components.

(45) In some examples, electrical sensing circuit **86** includes multiple sensing channels for acquiring cardiac electrical signals from multiple sensing vectors selected from electrodes **24**, **26**, **28** and **30** and housing **15**. Each sensing channel may be configured to amplify, filter and rectify the cardiac electrical signal received from selected electrodes coupled to the respective sensing channel to improve the signal quality for sensing cardiac events, e.g., P-waves and/or R-waves. Each sensing channel includes cardiac event detection circuitry for sensing cardiac events from the received cardiac electrical signal developed across the selected sensing electrode vector(s). For example, each sensing channel in sensing circuit **86** may include an input or pre-filter and amplifier for receiving a cardiac electrical signal from a respective sensing vector, an analog-to-digital converter, a post-amplifier and filter, and a rectifier to produce a digitized, rectified and amplified cardiac electrical signal that is passed to a cardiac event detector included in sensing circuit **86** and/or passed to control circuit **80** for performing signal analysis.

(46) The cardiac event detector may include a sense amplifier, comparator or other circuitry for comparing the rectified cardiac electrical signal to a cardiac event sensing threshold amplitude, such as an R-wave sensing threshold amplitude, which may be an auto-adjusting threshold. Sensing circuit **86** may produce a sensed cardiac event signal in response to a sensing threshold crossing.

The sensed cardiac events (or lack thereof), e.g., R-waves, are used for detecting cardiac rhythms and determining a need for therapy by control circuit **80**. In some examples, cardiac electrical signals such as sensed R-waves are used to detect capture of a pacing pulse delivered by ICD **14**. (47) Therapy delivery circuit **84** may include a low voltage (LV) therapy circuit **85** for delivering low voltage pacing pulses using an extra-cardiovascular pacing electrode vector selected from electrodes **24**, **26**, **28** and **30** and housing **15**. LV therapy circuit **85** may be configured to deliver low voltage pacing pulses, e.g., 8 V or less or 10 V or less in pulse amplitude. One or more capacitors included in the LV therapy circuit **85** are charged to a voltage according to a programmed pacing pulse amplitude by a LV charging circuit, which may include a state machine. The LV charging circuit may charge the capacitors to a multiple of the voltage of a battery included in power source **98** without requiring a transformer. At an appropriate time, the LV therapy circuit **85** couples the holding capacitor(s) to a pacing electrode vector, e.g., via a tip or output capacitor, to deliver a pacing pulse to the heart **8**. One example of a LV therapy circuit is described below in conjunction with FIG. 5.

(48) High voltage (HV) therapy circuit **83** includes one or more high voltage capacitors. When a shockable rhythm is detected, the HV capacitor(s) is(are) charged to a shock voltage amplitude by a HV charging circuit according to the programmed shock energy. The HV charging circuit **83** may include a transformer and be a processor-controlled charging circuit that is controlled by control circuit **80**. Control circuit **80** applies a signal to trigger discharge of the HV capacitor(s) upon detecting a feedback signal from therapy delivery circuit **84** that the HV capacitors have reached the shock voltage amplitude required to deliver the programmed shock energy. In this way, control circuit **80** controls operation of the HV therapy circuit **83** to deliver CV/DF shocks using defibrillation electrodes **24**, **26** and/or housing **15**.

(49) HV therapy circuit **83** may be used to deliver cardiac pacing pulses in some applications. In this case, the HV capacitor(s) is(are) charged to a much lower voltage than that used for delivering shock therapies but may be higher than the maximum available pulse voltage amplitude produced by the LV therapy circuit **85**. For example, the HV capacitor may be charged to 40 V or less, 30 V or less, or 20 V or less for producing extra-cardiovascular pacing pulses. In some examples, HV therapy circuit **83** is enabled to deliver cardiac pacing pulses by applying at least a minimum electrical current required to maintain switches included in HV therapy circuit **83** in a closed state as needed for coupling the HV capacitor(s) to a pacing electrode vector. Circuitry included in HV therapy circuit **83** is described in conjunction with FIG. 4 below.

(50) Compared to pacing pulses delivered by LV therapy circuit **85**, pulses delivered by HV therapy circuit **83** may have a higher voltage amplitude and/or relatively longer pulse width for delivering higher energy pacing pulses for capturing the heart. More current may be delivered using a low impedance pacing electrode vector, e.g., between electrodes **24** and **26**. Longer pulse width is attainable due to a higher capacitance (and consequently higher RC time constant) of the HV capacitor(s). The LV therapy circuit **85** may be capable of producing a maximum pulse voltage amplitude of up to and including 10 V. The maximum single-pulse pacing pulse width produced by LV therapy circuit **85** may be 2 ms. In some examples, LV therapy circuit **85** may be configured to produce composite pacing pulses comprising two or more individual pulses fused in time to deliver a cumulative composite pacing pulse energy that captures the heart. Techniques for delivering composite pacing pulses are generally disclosed in U.S. patent application Publication No. 2017/0157413 (Anderson, et al.) and U.S. patent application Publication No. 2017/0157399 (Anderson, et al.), both incorporated herein by reference in their entirety. The composite pacing pulse width may be 4 ms or higher, for example.

(51) The HV therapy circuit **83** may be capable of producing a pulse voltage amplitude of 10 V or more and may produce mono- or multi-phasic pulses having a relatively longer pacing pulse width, e.g., 10 ms or more, because of the higher capacitance of high voltage capacitors included in HV therapy circuit **83**. A typical HV pacing pulse width may be 10 ms; however an example range of

available pulse widths may be 2 ms to 20 ms. An example of a maximum voltage amplitude that may be used for delivering high voltage pacing pulses may be 40 V. When a relatively higher pacing pulse voltage amplitude is tolerable by the patient, e.g., more than 10 V, a relatively shorter pacing pulse width, e.g., 2 to 5 ms, may be used during an extra-cardiovascular pacing output configuration. However, a longer pacing pulse width may be used as needed, e.g., a 10 V, 20 ms pacing pulse, to capture the patient's heart without causing pain or discomfort.

(52) For the sake of comparison, the HV capacitor(s) of the HV therapy circuit **83** may be charged to an effective voltage greater than 100 V for delivering a CV/DF shock. For example, two or three HV capacitors may be provided in series having an effective capacitance of 148 to 155 microfarads in HV therapy circuit **83**. These series capacitors may be charged to develop 100 to 800 V for the series combination in order to deliver shocks having a pulse energy of 5 Joules or more, and more typically 20 Joules or more.

(53) In contrast, pacing pulses delivered by the HV therapy circuit **83** may have a pulse energy less than 1 Joule and even in the millijoule range or tenths of millijoules range depending on the pacing electrode impedance. For instance, a pacing pulse generated by HV therapy circuit **83** having a 10 V amplitude and 20 ms pulse width delivered using a pacing electrode vector between defibrillation electrodes **24** and **26**, having an impedance in the range of 20 to 200 ohms, may have a delivered energy of 5 to 7 millijoules. When a relatively shorter pulse width is used, e.g., down to 2 ms, the pacing pulse delivered by HV therapy circuit **83** using defibrillation electrodes **24** and **26** may be as low as 1 millijoule. Pacing pulses delivered by HV therapy circuit **83** are expected to have a pacing voltage amplitude that is less than 100 V, and typically not more than 40 V, and deliver at least 1 millijoule but less than 1 Joule of energy. The delivered energy for a given pacing voltage amplitude will vary depending on the pulse width and pacing electrode vector impedance.

(54) If a pace/sense electrode **28** or **30** is included in the pacing electrode vector, resulting in a relatively higher impedance, e.g., in the 400 to 1000 ohm range, the pacing pulse energy delivered may be in the range of 2 to 5 millijoules. HV therapy circuit **83** may deliver more current via a lower impedance pacing electrode vector, e.g., between defibrillation electrodes **24** and **26**, than the current delivered by LV therapy circuit **85** via a pacing electrode vector including a pace/sense electrode **28** and **30** (relatively higher impedance) even when the pacing voltage amplitude is the same.

(55) Composite pacing pulses, delivered by the LV therapy circuit **85**, having an 8 V amplitude and 8 ms pulse width may be in the range of 0.5 to 1.3 millijoules for the range of pacing loads given in the preceding example. Extra-cardiovascular, single-pulse pacing pulses delivered by LV therapy circuit **83** that are 8V in amplitude and 2 ms in pulse width may be in the range of 0.2 to 0.3 millijoules for pacing loads of 400 to 1000 ohms. In contrast, pacing pulses delivered using endocardial electrodes or epicardial electrodes may be on the order of microjoules, e.g., 2 microjoules to 5 microjoules for a typical endocardial pacing pulse that is 2V in amplitude, 0.5 ms in pulse width and applied across a pacing electrode vector impedance of 400 to 1000 ohms. As such, even though LV therapy circuit **83** is referred to herein as a “low voltage” therapy circuit, cardiac pacing delivered by LV therapy circuit **83** may still be considered a high voltage pacing application relative to very low voltage pacing applications that utilize endocardial or epicardial electrodes because LV therapy circuit **85** is configured to deliver pacing pulses via extra-cardiovascular electrodes. As will be described below, control circuit **80** may control therapy circuit **84** to deliver cardiac pacing pulses using the charge balancing techniques disclosed herein. The cardiac pacing pulses may be generated by HV therapy circuit **83** or LV therapy circuit **85**.

(56) In some instances, control circuit **80** may control impedance measurement circuit **90** to determine the impedance of a pacing electrode vector. Impedance measurement circuit **90** may be electrically coupled to the available electrodes **24**, **26**, **28**, **30** and housing **15** for performing impedance measurements of one or more candidate pacing electrode vectors. Control circuit **80** may control impedance measurement circuit **90** to perform impedance measurements by passing a

signal to impedance measurement circuit **90** to initiate an impedance measurement of a pacing electrode vector. Impedance measurement circuit **90** is configured to apply a drive or excitation current across a pacing electrode vector and determine the resulting voltage. The voltage signal may be used directly as the impedance measurement or impedance may be determined from the applied current and the measured voltage. The impedance measurement may be passed to control circuit **80**.

(57) Control circuit **80** may use impedance measurements from impedance measurement circuit **90** to control delivery of pacing pulses using the charge balancing techniques disclosed herein. For instance, the pacing vector impedance may be used in selecting how often a modified pacing pulse is injected in a series of alternating polarity pacing pulses to provide charge balancing. As described below, the pacing electrode vector impedance may be used as an input to an algorithm for selecting the modification of an nth pacing pulse in a series of alternating polarity pacing pulses or used to determine the modification of an nth pacing pulse from a lookup table stored in memory **82** based on multiple impedance ranges.

(58) Control parameters utilized by control circuit **80** for detecting cardiac rhythms and delivering electrical stimulation therapies and tachyarrhythmia induction pulses, including charge balancing pacing pulse modifications, may be programmed into memory **82** via telemetry circuit **88**.

Telemetry circuit **88** includes a transceiver and antenna for communicating with external device **40** (shown in FIG. 1A) using RF communication as described above. Under the control of control circuit **80**, telemetry circuit **88** may receive downlink telemetry from and send uplink telemetry to external device **40**. In some cases, telemetry circuit **88** may be used to transmit and receive communication signals to/from another medical device implanted in patient **12**.

(59) FIG. 4 is a schematic diagram **200** of HV therapy circuit **83** coupled to a processor and HV therapy control circuit **230**. HV therapy circuit **83** includes a HV charging circuit **240** and a HV charge storage and output circuit **202**. Processor and HV therapy control circuit **230** may be included in control circuit **80** for controlling HV charging circuit **240** and HV charge storage and output circuit **202**.

(60) HV charge storage and output circuit **202** includes a HV capacitor **210** coupled to switching circuitry **204** via a pulse control switch **206** for electrically coupling the HV capacitor **210** to electrodes **24** and **26** and/or housing **15** to deliver a desired HV electrical stimulation pulse to the patient's heart **8**. HV capacitor **210** is shown as a single capacitor, but it is recognized that a bank of two or more capacitors or other energy storage devices may be used to store energy for producing electrical signals delivered to heart **8**. In one example, HV capacitor **210** is a series of three capacitors having an effective capacitance of 148 microfarads, 155 microfarads, or other selected capacitance. In contrast, holding capacitors that are included in LV therapy circuit **85** that are charged to a multiple of the battery voltage by a state machine may have a capacitance of up to 6 microfarads, up to 10 microfarads, up to 20 microfarads or other selected capacitance, but all have a capacitance significantly less than the effective capacitance of HV capacitor **210**. The LV therapy circuit **85** has a lower breakdown voltage than the HV therapy circuit **83**, allowing the HV capacitor **210** to be charged to the shock voltage amplitude, e.g., 100 V or more, required for delivering CV/DF shocks.

(61) Switching circuitry **204** may be in the form of an H-bridge including switches **212a-212c** and **214a-214c** that are controlled by signals from processor and HV control circuit **230**. Switches **212a-212c** and **214a-214c** may be implemented as silicon-controlled rectifiers (SCRs), insulated-gate bipolar transistors (IGBTs), metal-oxide-semiconductor field-effect transistors (MOSFETs), and/or other switching circuit components or combinations thereof.

(62) When control circuit **80** determines that delivery of an electrical stimulation pulse from HV therapy circuit **83** is needed, switching circuitry **204** is controlled by signals from processor and HV therapy control circuit **230** to electrically couple HV capacitor **210** to a therapy delivery vector to discharge capacitor **210** across the vector selected from electrodes **24**, **26** and/or housing **15**. The

selected electrodes **24**, **26** and/or housing **15** are coupled to HV capacitor **210** by opening (i.e., turning off or disabling) and closing (i.e., turning on or enabling) the appropriate switches of switching circuitry **204** to pass a desired electrical signal to the therapy delivery electrode vector. While only electrodes **24**, **26** and housing **15** are indicated as being coupled to switching circuitry **204**, it is to be understood that pace/sense electrodes **28** and **30** may be coupled to switching circuitry **204** and available for use in a pacing electrode vector.

(63) When control circuit **80** determines that a shock therapy is needed based on a detected heart rhythm, e.g., VT or VF, the electrical signal delivered by HV therapy circuit **83** may be a monophasic, biphasic or other shaped CV/DF shock pulse for terminating the ventricular tachyarrhythmia. When control circuit **80** determines that a pacing therapy is needed based on a detected heart rhythm or a pacing escape interval expiring, the electrical signals delivered by HV therapy circuit **83** include alternating polarity pulses with every nth pulse being modified to provide charge balancing. Modification of every nth pulse may include modification of the polarity, pulse amplitude, and/or pulse width compared to an expected, unmodified pattern of the alternating polarity pulses.

(64) The pacing pulse may be terminated when the pulse voltage amplitude has decayed according to a programmed “tilt.” Tilt is the percentage of the leading voltage amplitude that the pulse has decayed to. For example, the pacing pulse may be terminated at a tilt of 50%, when the pulse amplitude has decayed to 50% of the leading voltage amplitude. If the programmed tilt is 20%, the pacing pulse may be terminated when the pulse amplitude has decayed by 20%, i.e., to 80% of the leading voltage amplitude. In other examples, the pacing pulses delivered by HV pacing circuit **83** are terminated according to a predetermined pulse width.

(65) Processor and HV therapy control circuit **230** controls charging of capacitor **210** to a programmed pacing pulse voltage amplitude or a modified, charge-balancing pacing pulse voltage amplitude. Switches **212a-212c** and **214a-214c** are controlled to be open or closed by processor and HV therapy control circuit **230** at the appropriate times for delivering a monophasic, biphasic or other desired pacing pulse by discharging capacitor **210** across the pacing load presented by heart **8** and a selected pacing electrode vector, e.g., electrodes **24** and **26**. The capacitor **210** is coupled across the selected pacing electrode vector for a programmed or modified pacing pulse width or according to a programmed tilt.

(66) Processor and HV therapy control **230** may control HV therapy circuit **83** to deliver cardiac pacing pulses by controlling HV therapy circuit **83** to deliver alternating polarity pacing pulses. Inherent limitations of the therapy delivery circuit **84** may result in charge imbalanced alternating polarity pacing pulses. For example, a net 10% charge difference may exist between a positive-going pulse and a negative going pulse that are controlled by therapy delivery circuit **84** according to the same programmed pulse amplitude (though opposite in polarity) and the same programmed pulse width (or tilt). Over time this small charge difference accumulates and may lead to electrode corrosion. In this example, assuming that each positive polarity pulse delivers approximately 10% greater charge than each negative polarity pulse, control circuit **80** may deliver ten alternating polarity pulses (+--+--+ . . .) and modify the polarity of every eleventh pulse to be a negative polarity pulse (instead of the expected positive polarity) such that two consecutive negative polarity pulses are delivered, to balance the net positive charge accumulated over the preceding ten, alternating polarity pulses. In other examples, processor and HV therapy control **230** may control HV therapy circuit **83** to deliver every nth pacing pulse with a modified pulse amplitude and/or modified pulse width (compared to the programmed pacing pulse amplitude and pulse width of the preceding alternating polarity pulses) to balance charge accumulated in one or more preceding pacing pulses. The examples presented herein generally refer to the nth pulse of a series of n pulses being the modified, charge balancing pulse. It is contemplated, however, that the modified charge balancing pulse is not necessarily the last pulse of a series of n pulses. Any pulse in a series of n pulses may be the modified, charge balancing pulse that reduces or eliminates the net charge

accumulated over the other $n-1$ pulses in the series of n pulses.

(67) Before the first pulse, the HV capacitor **210** may be charged to the programmed pacing pulse voltage amplitude. HV capacitor **210** is recharged between consecutive pulses to the programmed pacing pulse voltage amplitude. HV charging circuit **240** is powered by power source **98** (FIG. 3). HV charging circuit **240** includes a transformer **242** to step up the battery voltage of power source **98** in order to achieve charging of capacitor **210** to a voltage that is much greater than the battery voltage. Charging of capacitor **210** by HV charging circuit **240** is performed under the control of processor and HV therapy control **230**, which receives feedback signals from HV charge storage and output circuit **202** to determine when capacitor **210** is charged to a programmed voltage. A charge completion signal is passed to HV charging circuit **240** to terminate charging by processor and HV therapy control circuit **230**. One example of a high voltage charging circuit and its operation is generally disclosed in U.S. Pat. No. 8,195,291 (Norton, et al.), incorporated herein by reference in its entirety.

(68) Switches **212a-212c** and switches **214a-214c** may require a minimum current flow to hold them closed (i.e., ON or enabled) for passing current as HV capacitor **210** is discharged. The minimum current to enable (close) switches **214a-214c** may be approximately 10 milliamps or less than 10 milliamps depending on the pacing load impedance and other conditions. The electrical current passing through the enabled ones switches **212a-212c** and **214a-214c** may fall below the minimum current required to keep the enabled switches closed as capacitor **210** is discharged across a selected pacing electrode vector. If the current passing through a respective switch falls below the minimum current required to keep the switch closed, the switch may open (or become disabled) causing premature truncation of the pacing pulse, which could result in loss of capture and/or inadequate charge balancing. As such, a minimum pacing pulse voltage amplitude may be set for delivering pulses from HV therapy circuit **83** in order to reduce the likelihood of the electrical current falling below the minimum current required to maintain a stable state of enabled switches of switching circuitry **204** during a programmed pacing pulse width (or until a programmed tilt has been reached).

(69) In other examples, HV charge storage and output circuit **202** may include an optional shunt resistance **250**, **252** in parallel to the pacing load to maintain a minimum current through the switching circuitry **204**. Shunt resistance **250** is shown in parallel to the pacing load shown schematically as heart **8** when electrodes **24** and **26** are selected as the anode and cathode (or cathode and anode, respectively) of the pacing electrode vector. It is recognized that a shunt resistance may be provided in parallel to the pacing load for any selected pacing electrode vector, for example shunt resistance **252** is shown schematically if the pacing electrode vector includes electrode **26** and housing **15**. Likewise a shunt resistance may be provided in parallel to the pacing load when the pacing electrode vector includes electrode **24** and housing **15**. The shunt resistance **250** or **252** may be a variable resistance that is set to match the pacing electrode vector impedance so that the load across heart **8** using a selected pacing electrode vector matches the shunt resistance. In this way, current through the switching circuitry **204** may be maintained at or above a minimum current required to maintain a stable state of enabled switches of switching circuitry **204** during the pacing pulse. Other examples of the use of a shunt resistance in switching circuitry **204** are generally disclosed in the above-incorporated U.S. patent application Ser. No. 15/425,169.

(70) In some examples, the pacing electrode vector coupled to HV capacitor **210** via switching circuitry **204** may include electrodes **24**, **26**, **28** and/or **30** carried by lead **16**. Housing **15** may be unused for cardiac pacing pulse delivery by holding switches **212c** and **214c** open. Depending on the implant location of ICD **14** and lead **16** and the resulting electrical stimulation delivery vector between the housing **15** and an electrode **24**, **26**, **28** and/or **30**, greater recruitment of skeletal muscle may occur when housing **15** is included in the pacing electrode vector. A larger volume of skeletal muscle tissue may lie along a vector extending between the distal portion **25** of lead **16** and housing **15** than along a vector extending between the two electrodes carried by lead distal portion

25. In other electrode configurations and implant locations, however, the electrodes (which may include ICD housing **15**) used to deliver extra-cardiovascular pacing pulses by HV therapy circuit **83** may be selected to provide a pacing electrode vector that minimizes the volume of skeletal muscle included in the pacing electrode vector while directing sufficient energy to the heart **8** for capturing and pacing the heart.

(71) FIG. 5 is a conceptual diagram of LV therapy circuit **85** according to one example. LV therapy circuit **85** may include a capacitor selection and control circuit **504**, a LV charging circuit **514**, and a capacitor array **610**. Capacitor array **610** may include multiple holding capacitors **612**, **614**, **616** and **618** that can each be charged by LV charging circuit **514** to a programmed pacing pulse amplitude. The holding capacitors **612**, **614**, **616** and **618** are coupled to a respective output capacitor **632a-632d** (collectively **632**), **636**, or **638** via respective switches **622**, **624**, **626**, and **628** to deliver pacing pulses. Each of holding capacitors **612**, **614**, **616** and **618** has a capacitance that is less than the effective capacitance of high voltage capacitor **210** of HV therapy circuit **83**. For example each of holding capacitors **612**, **614**, **616** and **618** may have a capacitance of up to 6 microfarads, up to 10 microfarads, up to 20 microfarads or other selected capacitance, but all have a capacitance significantly less than the effective capacitance of high voltage capacitor **210**.

(72) Power source **98** (FIG. 3) may provide regulated power to LV charging circuit **514**. LV charging circuit **514** may be controlled by a state machine in capacitor selection and control circuit **504** to charge all or selected ones of holding capacitors **612**, **614**, **616** and **618** using a multiple of the battery voltage of power source **98**, e.g., four times the battery voltage. LV charging circuit **514** charges capacitors **612**, **614**, **616** and/or **618** as needed for delivering pacing pulses, which may be single pacing pulses (e.g., monophasic or biphasic) or composite pacing pulses during fused pacing. Composite pacing pulses include two or more individual pulses delivered successively so that the pulse energy of the successive individual pulses is effectively “fused” in time to deliver cumulative pacing pulse energy greater than the cardiac capture threshold. Fused pacing using composite pacing pulses delivered by a LV pacing circuit for extra-cardiovascular pacing is generally disclosed in U.S. Patent Application Publication No. 2017/0157399 (Anderson, et al.), incorporated herein by reference in its entirety.

(73) In some examples, the LV therapy circuit **85** includes three pacing channels **602**, **604** and **606**. Each pacing channel is capable of producing a single pacing pulse when a respective holding capacitor **612**, **616** or **618** is discharged across an output capacitor **632**, **636**, or **638**, respectively. Pacing channel **602** includes a back-up holding capacitor **614** that may be used for delivering back-up pacing pulses, e.g., when a single pulse pacing output is selected. Back-up holding capacitor **614** may be used to deliver an individual pulse of a composite pacing pulse when fused pacing is being delivered.

(74) Depending on the number of extra-cardiovascular electrodes coupled to ICD **14**, one or more channels may include multiple selectable output signal lines. For example, channel **602** is shown in this example to include multiple selectable pacing output signal lines **642a-642d** that may be selectively coupled to holding capacitor **612** and back-up holding capacitor **614** via closure of one or more of electrode selection switches **634a-634d**. For example, multiple electrodes carried by lead **16** may be coupled to pacing channel **602** and a pacing electrode vector may be selected from the multiple electrodes by closing certain ones of switches **634a-634d**.

(75) Pacing channels **604** and **606** are shown having single output signal lines **646** and **648** that are coupled to respective holding capacitors **616** and **618** via respective switches **626** and **628**. In other examples, all three pacing channels **602**, **604** and **606** may be provided with a single output signal line or with multiple output signal lines to enable selection of a pacing electrode vector from among multiple extra-cardiovascular electrodes coupled to ICD **14**, e.g., any of electrodes **24**, **26**, **28** or **30** of lead **16**.

(76) A single channel pacing output configuration may be enabled by control circuit **80** using any one of the pacing channels **602**, **604** and **606** to deliver single-pulse pacing pulses. The single-pulse

pacing may be delivered by discharging one of the holding capacitors **612**, **614**, **616** or **618** across a selected pacing electrode vector via a respective output capacitor **632**, **636** or **638** when a respective switch **622**, **624**, **626** or **628** is closed. The output line **642a**, **642b**, **642c**, or **642d** used to deliver pacing current from pacing channel **602** may be selected via a respective electrode selection switch **634a-634d**. The switches **622**, **624**, **626** or **628** that enable discharge of a holding capacitor **612**, **614**, **616**, or **618**, respectively, may be enabled by capacitor selection and control circuit **504** at the appropriate time when a pacing pulse is needed and maintained in an active, enabled state until the single pacing pulse width is expired.

(77) When control circuit **80** is configured to deliver fused pacing using LV therapy circuit **85**, the pacing channels **602**, **604** and **606** are tied together by switches **620a-620d** and **630** to enable individual pulses to be delivered across a selected pacing electrode vector from a single output signal line, e.g., line **646**. For example, control circuit **80** may enable the fused pacing pulse output by activating switches **620a-620b** and **630** to tie pacing output lines **642a-642d** and pacing output line **648** to pacing channel **604**. Control circuit **80** controls capacitor selection and control circuit **504** to enable pacing channel switches **622**, **624**, **626** and **628** (and at least one electrode selection switch **634a-634d** of pacing channel **602**) in a sequential manner to couple a respective holding capacitor **612**, **614**, **616** or **618** to output signal line **646** to deliver a sequence of fused, individual pulses to produce a composite pacing pulse.

(78) In various examples, depending on the particular pacing channel and lead and electrode configuration used with ICD **14**, some electrode selection switches shown in FIG. 5 may not be required. Furthermore it is recognized that less than four holding capacitors or more than four holding capacitors may be included in a capacitor array **610** for use in delivering a sequence of fused pacing pulses.

(79) Capacitor selection and control circuit **504** selects which holding capacitors **612**, **614**, **616** and **618** are coupled to output line **646** and in what sequence by controlling respective switches **622**, **624**, **626** and **628**. A sequence of pulses may be delivered to produce a composite pacing pulse by sequentially discharging holding capacitors **612**, **614**, **616** and **618** one at a time (or one combination at a time) across a respective output capacitor **632**, **636** and **638** by sequentially enabling or closing the respective switches **622**, **624**, **626** and **628**. For example, at least two of holding capacitors **612**, **614**, **616** and **618** are sequentially discharged to produce a composite pacing pulse produced by at least two fused individual pulses. Output line **646** may be electrically coupled to a pacing cathode electrode carried by lead **16** and a return anode electrode carried by lead **16** (or housing **15**) may be coupled to ground. The pacing cathode electrode and return anode electrode may correspond to electrodes **24** and **28** or any pacing electrode vector selected from electrodes **24**, **26**, **28** and **30** and/or housing **15** shown in FIGS. 1A-2B.

(80) In some examples, a fused pacing pulse is delivered by delivering two consecutive individual pulses from pacing channel **604** and **606** one at a time followed by a third, longer individual pulse delivered by pacing channel **602** by discharging both capacitors **612** and **614** simultaneously. The first two individual pulses may be 2.0 ms in pulse width and the third pulse may be 4.0 ms in pulse width for a composite pacing pulse width of 8 ms. The higher capacitance of the parallel capacitors **612** and **614** allows for the third individual pulse to be longer in pulse width while maintaining a pulse amplitude for the duration of the pulse width that successfully captures the heart. All three individual pulses are delivered via output line **646** because output configuration switches **620** and **630** are enabled for the fused pacing output configuration. In other examples, selected ones of holding capacitors **612**, **614**, **616**, and **618** are discharged sequentially, individual or in combination, to deliver a series of pulses closely together in time to form a composite pacing pulse.

(81) LV therapy circuit **85** may be controlled by control circuit **80** to deliver every nth pulse of a series of alternating polarity pacing pulses with a modified polarity, modified pulse amplitude, and/or modified pulse width to balance an unbalanced charge delivered over the preceding n-1 pacing pulses. The n-1 unmodified, alternating polarity pacing pulses may be delivered as

alternating polarity single-pulse pacing pulses or as alternating polarity composite pacing pulses comprising two or more fused pulses. Modification of the nth pulse may include modifying the expected polarity of the nth pulse in the pattern of alternating polarity pulses by delivering two consecutive pulses (the nth-1 pulse and the nth pulse) having the same polarity, either both positive or both negative. Modification of the nth pulse may include modifying the expected pulse amplitude and/or pulse width by increasing or decreasing the pulse amplitude and/or pulse width from the programmed pacing pulse amplitude and/or programmed pacing pulse width, respectively, which is used to control the preceding n-1 pulses.

(82) In the examples presented herein, the electrical stimulation pulses including opposing polarity pulses are voltage-controlled pulses such that the charge balancing nth pulse of a series of pulses may have a modified voltage amplitude. It is recognized however, depending on the medical device, the electrical stimulation pulses may be current-controlled pulses and, as such, the modification to the charge balancing nth pulse may be a modified current amplitude.

(83) FIG. 6 is a conceptual diagram 300 of a series of alternating polarity pacing pulses 302 including a charge balancing pulse 322. The series of n pulses 302 includes positive polarity pulses 304 and negative polarity pulses 306 that are delivered in alternation. The pulses are shown as monophasic pulses, but could be biphasic or other shaped pulses in other examples. The series of n pulses 302, including opposing polarity pulses, and series of pulses 350 and 380 shown in FIGS. 7 and 8, respectively, may represent a variety of electrical stimulation pulses that are delivered by a medical device to body tissue of a patient. In the illustrative examples described herein, the pulses may be cardiac pacing pulses which include opposing polarity pacing pulses and an nth charge balancing pacing pulse. In one example, control circuit 80 may control either HV therapy circuit 83 or LV therapy circuit 85 to deliver the alternating polarity pacing pulses 304 and 306 of the series of n pulses 302.

(84) The alternating polarity pulses 304 and 306 may be separated in time by a pacing pulse interval 320. The pacing pulse interval 320 may be a lower rate interval used to pace the heart when the intrinsic rate falls below the programmed lower rate. The lower rate interval may be adaptively controlled according to a sensor-indicated patient activity to provide rate responsive pacing in some examples. The pacing pulse interval 320 may alternatively be an ATP interval, a post-shock pacing interval, or other pacing interval set by control circuit 80 according to a programmed cardiac pacing therapy protocol. Pacing pulse interval 320 may stay constant during the series of n alternating pacing pulses 302 or may vary, according to the pacing protocol. In some examples, the pacing pulse interval 320 may be controlled by the expiration of an escape interval counter included in control circuit 80.

(85) In other examples, the pacing pulses 302 may represent ventricular pacing pulses being delivered in an atrial-tracking pacing mode or during CRT such that each pulse 304 and 306 is delivered upon the expiration of an AV pacing interval (not shown), in which case pulses 304 and 306 may be separated by variable intervals that depend on the atrial rate being tracked. In this case, the interval 320 between pulses does not necessarily represent a pacing escape interval. Furthermore, it is to be understood that consecutive alternating polarity pacing pulses may occur with extended periods of time (e.g., one or more unexpired pacing escape intervals) with intervening intrinsic cardiac events occurring in some cases. As such, one positive pacing pulse consecutively occurring with one negative pacing pulse in a pair of alternating polarity pulses are not required to occur on consecutive cardiac cycles. None, one or more intrinsic cardiac cycles may occur between the positive and negative polarity pacing pulses of a charge imbalanced pair of alternating polarity cardiac pacing pulses.

(86) Both of the positive polarity pulses 304 and the negative polarity pulses 306 may be delivered according to the same programmed pulse amplitude 310 and pulse width 312 (or tilt), which may be selected based on cardiac capture threshold testing. Inherent system limitations, however, may result in a net difference in the charge delivered during one pair of one positive pacing pulse 304

and one negative pacing pulse **306**. Depending on the frequency of pacing, even a small charge imbalance of each pair of alternating polarity pulses may result in an accumulation of imbalanced charge over time. Every n th pacing pulse **322** of series **302** is modified by reversing the polarity of the n th pulse in the alternating pattern of pacing pulses. The modified n th pacing pulse **322** is delivered having a polarity that is the opposite polarity of the charge imbalance of a pair of one positive and one negative pulse to balance or offset accumulated charge. The charge delivered during each positive pacing pulse **304** may be a known or approximated percentage greater than the charge delivered during each negative polarity pulse **306** such that the pair of pulses has a net positive charge delivered or imbalanced positive charge. In other cases, the negative polarity pulse **306** may deliver a greater charge than the positive pulse **304** resulting in a net negative charge accumulation that is balanced by a positive polarity charge balancing pulse.

(87) The value of n for controlling how often a modified, charge balancing pacing pulse is delivered may be based on the net charge difference between the positive and negative polarity pulses **304** and **306**. For instance n may be determined as twice the reciprocal of the charge imbalance plus one or $n=2*(1/c)+1$ where c is the percentage charge imbalance (in decimal form) of one alternating pair of one positive and one negative polarity cardiac pacing pulse. Every n th pulse **322** may be modified by delivering it having the opposite polarity of the imbalanced charge. To illustrate, if the positive polarity pulses **304** are approximately 10% greater in charge delivery than the negative polarity pulses **306**, a net positive charge equal to approximately 100% of the charge of one positive going pulse is accumulated after a series of 20 alternating pulses (10 positive-going pulses and 10 negative-going pulses). To balance the accumulated positive charge of 10% per alternating pacing pulse pair, control circuit **80** may determine that $n=21$ ($21=2(1/0.1)+1$). Every 21st pulse of the alternating polarity pulses is therefore modified by delivering it with a negative polarity to balance the accumulated imbalanced positive charge of the preceding 20 pulses. In this way, any net charge that accumulates over the preceding $n-1$ alternating pulses is substantially balanced by the n th pulse.

(88) The examples presented herein refer to a series of $n-1$ alternating polarity pulses followed by an n th pulse of the series being modified to provide charge balancing. It is contemplated, however, that the $n-1$ pulses may include any series of electrical stimulation pulses that results in a net charge accumulation that is offset by the n th pulse of the series. For instance, the $n-1$ pulses may include charge imbalanced, opposing polarity pulses but not necessarily in an alternating pattern of every other pulse being positive and every other pulse being negative. A series of pulses may include two, three or more consecutive positive polarity pulses followed by a like number of negative polarity pulses and that pattern may repeat until the n th modified, charge balancing pulse. To illustrate, the series of pulses including opposing polarity pulses may include two positive pulses followed by two negative pulses (+, +, -, -, +, +, -, -, +, + . . .), and the series may be terminated with the modified charge balancing pulse.

(89) While cardiac pacing pulses continuing after series **302** are not shown in FIG. 6, it is to be understood that series **302** may be n pacing pulses out of ongoing cardiac pacing pulses that are delivered according a programmed pacing therapy or pacing mode. For example, ICD **314** may be programmed to deliver cardiac pacing in VVI, VDI, or VDD pacing mode (with or without rate response) such that pacing may be occurring over an extended period time and series **302** represents n pacing pulses delivered during the ongoing pacing. During the ongoing pacing, every n th pulse may be modified by delivering it at the same polarity as the pulse polarity having an inherently lesser net charge. As a result, every $n-1$ and n th pulses during ongoing pacing may have the same polarity to balance a net charge accumulated over the $n-1$ pulses. Polarity alternation may resume with the next $n+1$ pulse being the opposite polarity of the n th pulse, and being the first pulse of the next series of n pulses ending with a modified n th pulse. For example, if every 5th pulse is modified to balance a net charge delivered during the preceding four alternating polarity pulses, the pacing pulse polarity may be represented by: +--+--+--+--+ . . . and so on. In

other examples, the series of pacing pulses **302** may represent all pulses delivered during an ATP or other pacing therapy in which the last pacing pulse may be the n th pulse **322** that is modified to balance the charge delivered over the preceding $n-1$ pulses of the pacing sequence.

(90) FIG. **7** is a diagram **350** of cardiac pacing pulses delivered according to another example technique for charge balancing. In this example every n th pulse **372** of each series of n pulses **352** is modified by adjusting the pacing pulse amplitude **374** of the n th pulse **352**. Each unmodified positive polarity pulse **354** and each unmodified negative going pulse **356** is delivered by therapy circuit **84** according to a programmed pacing pulse amplitude **360** and pulse width **362** (or tilt). Despite being delivered according to the programmed pacing pulse amplitude **360** and pulse width **362**, the alternating positive and negative pulses **354** and **356**, respectfully, may deliver an inherently different charge resulting in an imbalanced charge delivery after each pair of one positive pulse **354** and one negative pulse **356**. In the example shown, the n th pulse **372** of each series of n pulses **352** is modified by increasing the pulse amplitude **374** by a predetermined amount to balance a net positive charge accumulated during the preceding $n-1$ pulses. The increased pulse amplitude **374** is based on the inherently higher charge delivered with each positive polarity pulse **354** in this example.

(91) For instance, if each positive pulse **354** delivers a 10% higher charge than each negative pulse **356**, the modified negative polarity pulse **372** may have a modified pulse amplitude **374** that is adjusted to be greater than the unmodified pulse amplitude **360** to balance the 20% net positive charge imbalance of the preceding two pairs of one positive and one negative pulse. The value of the increased amplitude **374** may be determined based on bench testing or modeling of the charge imbalance between positive and negative pulses **354** and **356**. The amplitude adjustment made to the modified pulse amplitude may be dependent on the lead and electrode vector impedance and other electrical stimulation pulse output factors. In the example shown, $n=4$ so that every fourth pulse is delivered at an increased pulse amplitude. In other examples, every other pulse (e.g., every positive pulse or every negative pulse) may be delivered with a modified pulse amplitude so that the series of pacing pulses **352** include alternating polarity and alternating pulse amplitude pulses. In this case each pair of one positive polarity pulse **354** and one negative polarity pulse **356** is delivered as a series of $n=2$ pulses and in a way that minimizes any inherent charge imbalance.

(92) In some examples, control circuit **80** may be configured to establish how often a modified pulse is delivered and how much the pulse amplitude is adjusted in order to balance the preceding $n-1$ pulses based on the inherent charge difference between the positive and negative polarity pulses **354** and **356** and a maximum acceptable pulse amplitude of the modified, charge balancing pulse. If the charge imbalance of one pair of alternating polarity pulses is relatively high, n may be selected to have a relatively low value such that the modified n th pulse occurs relatively frequently with a reasonable and acceptable increase in pulse amplitude. If the charge imbalance of one pair of alternating polarity pulses is relatively low, a larger value of n may be selected so that the frequency of the modified, charge balancing pulse can be relatively lower, still with an acceptable higher pulse amplitude. An unacceptably high pulse amplitude may cause unintended capture of skeletal muscle or nerves in the vicinity of the pacing electrodes and cause patient discomfort.

(93) While the modified pulse **372** is shown in FIG. **7** to have an increased pulse amplitude compared to the unmodified alternating polarity pulses **354** and **352** to offset a charge imbalance of the opposite polarity of the modified pulse **372**, it is conceivable that a modified pulse in the series of n pulses could have a decreased pulse amplitude instead of an increased pulse amplitude. The decrease in amplitude may offset the inherently higher charge of the preceding pulses of the same polarity as the modified charge balancing pulse. The decreased pulse amplitude, however, may result in a loss of cardiac capture on the n th pulse, which may or may not be acceptable depending on the particular pacing application.

(94) FIG. **8** is a diagram **380** of another technique for delivering charge balancing cardiac pacing pulses. In FIG. **8**, the n th pulse **392** of each series **382** of alternating cardiac pacing pulses **304** and

306 is modified by adjusting the pulse width **384** from the unmodified pulse width **312**. The pulse width **384** may be increased or decreased to balance an accumulated net charge delivered over the preceding $n-1$ pulses. It is recognized that a decreased pulse width may result in a pacing loss of capture, which may or may not be acceptable. In the example shown, the n th pulse **392** is modified by increasing the pulse width **384** from the pulse width **312** of the unmodified pulses **304** and **306** to offset or balance a net positive charge accumulated during the delivery of the preceding $n-1$ pulses. The modified pacing pulse width may be determined based on a known inherent charge imbalance of each pair of unmodified, alternating polarity pacing pulses and the selected value of n . A selected value of n may have an upper limit based on a maximum pacing pulse width that is achievable by the RC time constant of the pacing circuit and acceptable in terms of overall delivered pacing pulse energy to the patient's heart. As such, if a relatively high charge imbalance exists, a lower value of n may be selected so that every n th pacing pulse is delivered with an acceptable and achievable increased pulse width **384** of the modified, charge balancing pulse **392**. (95) In the above examples, the positive polarity pulse is described as inherently delivering a greater charge than the negative polarity pulse such that after each pair of one positive and one negative pacing pulse, a net positive charge is delivered. The modified n th pulse in these examples is described as being a negative polarity pulse. It is recognized, however, that the modified, n th pulse may be a positive polarity pulse that is adjusted to balance a charge imbalance. The charge imbalance may be due to a net positive polarity charge or due to a net negative polarity charge since, in some cases, the negative polarity pulses may inherently deliver a greater charge than the positive polarity pulses.

(96) Each of the pulses shown in FIGS. **6-8** are cardiac pacing pulses that are intended to be therapeutic and therefore delivered at a pulse amplitude and width that is expected to cause cardiac capture and depolarization of the myocardial tissue. As such, each n th, charge balancing pacing pulse is also a therapeutic cardiac pacing pulse. In other examples, however, a modified n th pulse may be delivered during a physiological refractory period of the myocardium following the $n-1$ pulse such that the n th pulse is a non-therapeutic pulse that is delivered as a charge balancing pulse without the intention of causing cardiac capture. In other examples, the modification to the n th pulse may change the delivered pacing pulse energy to a level that is less than the pacing capture threshold. In this case, a loss of capture may occur, but loss of capture on one out of every n pacing pulses may be acceptable in some applications.

(97) FIG. **9** is a flow chart **400** of a method for delivering charge balancing cardiac pacing pulses according to one example. At block **402**, control circuit **80** establishes one or more charge balancing pacing pulse modifications to be performed during cardiac pacing. Control circuit **80** may control HV therapy circuit **83** or LV therapy circuit **85** to deliver extra-cardiovascular pacing pulses. Extra-cardiovascular pacing pulses generated by either the HV therapy circuit **83** or the LV therapy circuit **85** are relatively higher energy pacing pulses than pulses delivered by endocardial or epicardial electrodes. As such, control circuit **80** may establish a charge balancing pulse modification to be applied for use when HV therapy circuit **83** is being used to generate and deliver cardiac pacing pulses and the same or a different modification used when LV therapy circuit **85** is selected for generating and delivering cardiac pacing pulses. The charge balancing pacing pulse modification may be different when LV therapy circuit **85** is being used to generate and deliver cardiac pacing pulses than when HV therapy circuit **83** is being used since a different inherent charge imbalance may exist between the positive and negative polarity pulses generated by the two different circuits.

(98) In some examples, charge balancing pacing pulse modifications may be stored in a lookup table (LUT) in memory **82** for one or more pacing output conditions. A known inherent charge difference between positive and negative polarity pulses generated by therapy circuit **84** may be determined at the time of manufacture of the ICD **14**, e.g., through empirical or bench testing. The inherent charge difference may be determined for one or more pacing output conditions. For

example, a different charge imbalance may exist between positive and negative polarity pulses generated by HV therapy circuit **83** than the charge imbalance between positive and negative polarity pulses generated by LV therapy circuit **85**. Other pacing output conditions that may change the inherent charge difference between positive and negative polarity pulses may include the pacing electrode vector used to deliver the pulses, pacing electrode vector impedance, pacing pulse amplitude, pacing pulse width, and pacing rate, as examples and without limitation. An inherent charge difference and/or associated pacing pulse modification may be stored in memory **82** for each one of multiple pacing output conditions. The inherent charge difference(s) and/or corresponding pacing pulse modification(s) may be stored in one or more LUTs.

(99) Control circuit **80** may therefore establish the charge balancing modification at block **402** by retrieving the modification from a LUT stored in memory **82** for a given pacing output condition or combination of conditions. For instance, one LUT may be stored in memory **82** for use when HV therapy circuit **83** is used for delivering cardiac pacing pulses. The LUT may include multiple ranges of pacing electrode vector impedance paired with multiple ranges of pacing pulse amplitude. Each cell in the LUT may store the modification to be applied for a given combination of impedance and pacing pulse amplitude when the HV therapy circuit **83** is selected for generating and delivering the pacing pulses. Another LUT may be stored in memory **82** for use when the LV therapy circuit **85** is selected for delivering cardiac pacing pulses. This LV therapy circuit LUT may include the same or a different combination of pacing output conditions than the HV therapy circuit LUT since different conditions may have a greater or lesser impact on the inherent charge imbalance produced by the two different therapy circuits **83** and **85**. The LV therapy circuit LUT may include multiple ranges of pacing electrode vector impedance paired with multiple ranges of pacing pulse width, as an example.

(100) Each cell in a charge balancing modification LUT may at least define the value of n , where n is the number of the modified pulse in a series of consecutive, alternating-polarity cardiac pacing pulses. Each cell in the LUT may also include one or more pacing pulse modifications to be applied to the n th pacing pulse. As described above in conjunction with FIGS. **6-8**, the modifications may include polarity reversal, pulse amplitude adjustment, pulse width adjustment or any combination thereof.

(101) In other examples, a LUT stored in memory **82** may store the inherent charge imbalance of a pair of alternating polarity pacing pulses for a given pacing output condition or combination of pacing output conditions. Control circuit **80** may be configured to determine the charge balancing modification at block **402** based on an inherent charge imbalance stored in memory **82**. For example, one or more LUTs may be stored in memory **82** that list the inherent charge imbalance for the HV therapy circuit **83** for one or more output conditions such as pacing electrode vector, pacing impedance, pacing rate, pacing pulse amplitude and/or pacing pulse width. Likewise, one or more LUTs may store the inherent charge imbalance for the LV therapy circuit **83** for one or more output conditions. Control circuit **80** may retrieve the charge imbalance for a given pacing output configuration or condition and determine the pacing modification to be applied to every n th cardiac pacing pulse. In some cases, control circuit **80** only determines the value of n to control how often a charge balancing pacing pulse is delivered having a predetermined modification. The predetermined modification may be a reversal of polarity, a fixed pulse amplitude change and/or a fixed pulse width change.

(102) When the charge balancing pulse is modified by switching its polarity, as described in conjunction with FIG. **6**, establishing how often the modified pulse polarity is applied, e.g., every n pulses, may be the only parameter that control circuit **80** needs to determine at block **402** to establish the charge balancing modification based on a stored inherent charge imbalance. For example, control circuit may determine a value of n based on the imbalanced charge of each pair of one positive polarity pulse and one negative polarity pulse as described in conjunction with FIG. **6**.

(103) In examples that include adjusting the pulse amplitude and/or pulse width of the modified

pulse, establishing the charge balancing modification at block **402** may additionally or alternatively include determining a pulse amplitude and/or pulse width adjustment applied to every nth pulse based on a stored, inherent charge imbalance. In some examples, n may be fixed such that using a known charge imbalance between positive and negative polarity pulses, control circuit may calculate an appropriate polarity, amplitude and/or pulse width adjustment that should be applied to the nth pulse to balance charge accumulated during the preceding n-1 pulses. Other examples of methods for establishing the charge balancing modification are described below in conjunction with FIG. **11**.

(104) Control circuit **80** may control the timing of pacing pulses delivered by therapy delivery circuit **84** by setting pacing escape interval timers or counters in accordance with a pacing mode or therapy. The pacing escape intervals may be programmed values stored in memory **82** (e.g., a fixed lower rate interval or an atrioventricular pacing interval), or automatically adjusted pacing intervals (e.g., a rate responsive temporary lower rate interval). Upon expiration of a pacing interval at block **404**, control circuit **80** determines if the scheduled pacing pulse is the nth pulse of a series of consecutive alternating pacing pulses at block **406**. For example, control circuit **80** may include a counter that counts each delivered pacing pulse up to (or down from) n-1 delivered pacing pulses so that the nth pulse is delivered as the modified, charge balancing pulse at block **408**. If the scheduled pacing pulse is not the nth pulse, as determined at block **406**, control circuit **80** controls therapy delivery circuit to deliver an unmodified cardiac pacing pulse at block **410** according to the programmed pacing pulse amplitude, width and alternating polarity.

(105) Once the charge balancing modification is established at block **402**, no adjustments to the modification may be required in some examples. In other examples, control circuit **80** may determine that a change in pacing output has occurred at block **412**. One example of a change in pacing output is a change in pacing electrode vector impedance. Control circuit **80** may control impedance measurement circuit **90** to periodically monitor pacing electrode vector impedance (and/or in response to detecting loss of capture or selecting a new pacing electrode vector). If a change in impedance of the pacing electrode vector is detected (which may be due to a change in the selected pacing electrode vector), the charge imbalance between positive and negative polarity pulses may change. Other changes in pacing output detected at block **412** may include a change in the selection of the HV therapy circuit **83** or the LV therapy circuit **85** for generating the pacing pulses, a different pacing electrode vector selected for delivering pacing pulses, an adjustment of the pacing pulse amplitude, an adjustment of the pacing pulse width, and/or adjustment of a pacing mode (e.g., resulting in increased or decreased pacing rate or frequency).

(106) Control circuit **80** may be configured to detect or identify a change in a pacing output condition that is associated with a change in the charge imbalance between positive and negative polarity pulses, warranting a change in the charge balancing modification established at block **402**. In response to detecting a change in a pacing output condition that is associated with a change in charge imbalance, control circuit **80** may return to block **402** to select or determine a different charge balancing modification to account for a change in the charge imbalance of each pair of one positive and one negative polarity pacing pulse. Control circuit **80** may re-establish the charge balancing modification by retrieving data from LUTs stored in memory **82** as described above. If a change in pacing output is not detected at block **412**, control circuit **80** continues to count alternating polarity pacing pulses (blocks **404** and **406**) and controlling therapy delivery circuit **84** to deliver each nth pulse according to the established charge balancing modification at block **408**.

(107) FIG. **10** is a conceptual diagram of one example of a LUT **450** that may be stored in memory **82** for control circuit **80** to retrieve charge balancing pacing pulse modifications from. In this example, a pacing output condition is shown in the first column **452** as the pacing electrode vector impedance Z. Three ranges of impedance are shown in the first column **452**, but any desired number of pacing electrode vector impedances may be included in LUT **450**. Other examples of a pacing output condition that may be listed in the rows of the first column **452** may include but are

not limited to ranges of the unmodified pacing pulse amplitude, ranges of the unmodified pacing pulse width, or ranges of the pacing rate.

(108) A different pacing condition is shown across the first row of LUT **450**. In this example, a HV therapy circuit output configuration **454** and a LV therapy circuit output configuration **456** may be included in LUT **450**. In other examples, separate LUTs may be provided for each of the HV therapy circuit and the LV therapy circuit and may be one-dimensional or multi-dimensional tables which store a parameter used by control circuit **80** for establishing a charge balancing modification for a given set of pacing output conditions. In the example shown, control circuit **80** may be configured to determine the pacing electrode vector impedance (using impedance measurement circuit **90**) and the selected HV or LV therapy circuit for delivering pacing pulses and retrieve from a corresponding cell **458** the number *n* of the charge balancing modified pacing pulse. The *n*th pulse may be modified by reversing its polarity in the pattern of alternating polarity pulses.

(109) In other examples, each cell **458** of LUT **450** may indicate a pulse amplitude and/or pulse width modification to apply to every *n*th pacing pulse, where *n* is a fixed, predetermined value. In still other examples, each cell **458** may include a value of *n*, a pulse amplitude adjustment, a pulse width adjustment and an indication of whether the polarity should be reversed from the normal alternating order or any combination thereof. The stored pulse amplitude adjustment may range from 0 up to a maximum amplitude adjustment, e.g., up to a 1 to 5 V increase from the unmodified pacing pulse amplitude. The stored pulse width adjustment may range from 0 up to a maximum pulse width adjustment, e.g., up to a 5 to 10 ms increase from the unmodified pacing pulse width. One or more LUTs may be stored in memory **82** for enabling control circuit **80** to retrieve charge balancing pacing pulse modifications to be applied for a determined combination of pacing output conditions.

(110) In other examples, the LUT **450** may store in each cell **458** the net charge imbalance of one pair of one positive polarity pacing pulse and one negative polarity pacing pulse for a given combination of impedance and the selected therapy circuit, either HV therapy circuit **83** or LV therapy circuit **85**. Control circuit **80** may perform a computation or algorithm for determining the charge balancing modification based on the retrieved net charge imbalance. For example, control circuit **80** may determine the value of *n* based on the net charge imbalance as described in conjunction with FIG. **6**. Other mathematical functions modeling the relationship between pacing output conditions may be implemented in a processor of control circuit **80** for determining the charge balancing modification required to balance the retrieved net charge imbalance within predefined limits of pacing pulse amplitude and pacing pulse width.

(111) FIG. **11** is a flow chart **500** of a method for establishing a modification of a charge balancing pacing pulse by control circuit **80** of ICD **314** according to one example. The process of flow chart **500** may be performed at block **402** of FIG. **9** or may be performed for generating a LUT, such as LUT **450** of FIG. **10**. Control circuit **80** may be configured to perform an algorithm for determining a value of *n* (where *n* is the number of the modified pacing pulse in a series of consecutive pacing pulses), the polarity, the pulse amplitude and/or the pulse width of every modified *n*th pacing pulse. In order to determine the modification(s) of the charge balancing pacing pulse, control circuit **80** may determine one or more pacing output conditions that may change the inherent charge imbalance between a pair of one positive and one negative polarity pacing pulses.

(112) Control circuit **80** may select the pacing output configuration at block **501**. The pacing output configuration may be selected by selecting a therapy delivery output circuit and/or the pacing electrode vector. For instance, control circuit **80** may select the HV therapy circuit **83** or the LV therapy circuit **85** for delivering the pacing pulses via a selected pacing electrode vector. The LV therapy circuit **85** may be selected to deliver single-pulse pacing pulses or composite pacing pulses comprising multiple fused pulses delivered rapidly together.

(113) Control circuit **80** may determine the impedance of the selected pacing electrode vector at block **502** by controlling impedance measuring circuit **90**. Impedance may be determined by

delivering a drive current signal across a selected pacing electrode vector and determining a resulting voltage developed across the electrode vector. At block **505**, control circuit **80** may determine the pacing pulse amplitude and/or pacing pulse width of the cardiac pacing pulses. This determination may be made by performing a cardiac pacing threshold test for establishing the pacing pulse amplitude (or width) at a safety margin above the threshold pulse amplitude (or width). In other examples, the pacing pulse amplitude and width may be retrieved from memory **82** as stored pacing control parameters. While the determinations of pacing output configuration, electrode vector impedance and pulse amplitude and pulse width are shown as separate sequential steps in FIG. **11**, it is to be understood that these determinations may be performed in a semi-simultaneous or integrated manner since the pacing output configuration selected at block **501** may depend on the determined pacing electrode vector impedance and the pacing pulse amplitude and pacing pulse width will depend on the capture threshold determined for a given pacing electrode vector.

(114) At block **506**, control circuit **80** may determine an expected or historical pacing burden or rate. The expected pacing rate may be based on a programmed lower rate and/or pacing mode. For example, an expected average pacing rate during a rate responsive VVIR pacing mode may be higher than the expected average pacing rate during a non-rate responsive VVI pacing mode with a low programmed lower rate such as 40 pulses per minute or during VDD pacing with minimized ventricular pacing.

(115) At block **510**, control circuit **80** may determine an expected accumulated charge imbalance after each pair of one positive and one negative polarity pacing pulse based on the impedance, pacing pulse amplitude and width, expected pacing rate, and output configuration. In some examples, control circuit **80** may utilize the relationship $Q=V*PW/Z$ for determining Q as the delivered charge of a pacing pulse having a pulse amplitude V (in volts) and pulse width PW in seconds, using a pacing electrode vector having an impedance Z . In another example, control circuit **80** may measure the current I (in Amperes) delivered during a pacing pulse having pulse width PW (in seconds) and utilize the relationship $Q=I*PW$ for determining the charge Q delivered during the pacing pulse. The charge delivered for a set of pacing output conditions may be determined for a positive polarity pulse and for a negative polarity pulse for determining a net charge imbalance of the pair of one positive and one negative polarity pulse.

(116) One or more pacing condition inputs may be used to model or estimate the resulting charge imbalance during empirical testing. For instance a mathematical model of the net charge imbalance for one or more independent pacing output conditions may be generated during bench testing of ICD **314**. Control circuit **80** may provide the input pacing conditions to the model (stored in memory **82**) to determine the predicted charge imbalance of each pair of alternating polarity pulses. The predicted net charge imbalance of one pair of opposite polarity pulses may be output from the mathematical model after inputting the conditions determined at blocks **501** through **506**. Based on the predicted net charge imbalance of each pair of opposite polarity pacing pulses, control circuit **80** may determine the charge balancing modification to apply on every n th pacing pulse, which may include reversing the polarity, adjusting pulse amplitude and/or adjusting pulse width. Additionally or alternatively, control circuit **80** may be configured to determine n , the number of pacing pulses delivered in the series of alternating polarity pacing pulses ending with the modified charge balancing pulse and may apply a default modification, such as polarity reversal, to the n th pacing pulse. For example, if a net charge imbalance of 5% is determined based on the pacing output configuration and conditions determined at blocks **502** through **508**, control circuit **80** may determine n to be 41. The 41st pulse may be delivered with the opposite polarity of the charge imbalance of one pair of alternating polarity pulses to balance charge accumulated during the preceding 40 pacing pulses. Other examples of pacing pulse modification that control circuit **80** may determine at block **510** have been described above.

(117) While the examples described above include modifying every n th pulse using the same

modification(s) of polarity reversal, pulse amplitude and/or pulse width, it is further contemplated that the charge balancing cardiac pacing techniques disclosed herein may include modifying more than one pulse in a series of pulses, e.g., modifying every nth pulse using a first modification and modifying every mth pulse using a second modification different than the first modification. For instance, every nth pulse may be delivered with a modified pulse amplitude and every mth pulse may be delivered with a reversed polarity from the expected alternating polarity pattern. In one illustrative example, every 21st pulse is modified according to a first modification and every 201.sup.st pulse is modified according to a second modification. As such, in some examples, control circuit **80** may establish more than one charge balancing modification at block **510**. A first modification of every nth pulse and a second modification of every mth pulse may be determined. (118) The modification of every mth pulse may minimize any residual charge imbalance that remains after delivering a preceding number of series of n pulses. For instance, if positive pulses deliver a 10% higher charge than negative pulses, the modified nth pacing pulse may be the 21.sup.st pacing pulse delivered at negative polarity. This nth pulse may not balance 100% of the positive charge accumulated over the preceding 20 pulses, however. For example, the one negative pulse may balance 90% of the positive charge accumulated over the 20 preceding pulses since it delivers 10% less charge than the positive-going pulses. Therefore, every 201.sup.st pulse may be delivered having a negative polarity to minimize residual charge after every twenty, modified nth pulses.

(119) In such examples, m may have a higher value than the value of n. In other examples, m and n may be equal but are staggered such that in a series of pacing pulses the mth and nth pulses are spaced apart and may have different charge balancing modifications. For instance, if m and n are both equal to 5, in a series of 5 pulses the first two pulses may be unmodified (u), the third pulse may be modified according to the first modification (n), the fourth pulse may be unmodified and the fifth pulse may be modified according to the second modification (m) and this pattern of five pulses may repeat (u u n u m u u n u m . . .). The second modification of every mth pulse may further reduce any charge imbalance that is not eliminated by the first modification of every nth pulse.

(120) It should be understood that, depending on the example, certain acts or events of any of the methods described herein can be performed in a different sequence, may be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the method). Moreover, in certain examples, acts or events may be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors, rather than sequentially. In addition, while certain aspects of this disclosure are described as being performed by a single device, circuit or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or circuits associated with, for example, one or more medical devices.

(121) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media (e.g., RAM, ROM, EEPROM, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer).

(122) Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term "processor," as used herein may refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described herein. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(123) Thus, a medical device system has been presented in the foregoing description with reference to specific examples. It is to be understood that various aspects disclosed herein may be combined in different combinations than the specific combinations presented in the accompanying drawings. It is appreciated that various modifications to the referenced examples may be made without departing from the scope of the disclosure and the following claims.

Claims

1. A medical device comprising: a therapy delivery circuit configured to generate electrical stimulation pulses; and a control circuit configured to control the therapy delivery circuit to: generate the electrical stimulation pulses as alternating polarity pulses comprising $n-1$ alternating polarity therapeutic pulses that deliver a net electrical charge and an n th pulse, wherein the $n-1$ alternating polarity therapeutic pulses include at least one pair of opposite polarity therapeutic pulses such that n is at least 3; and modify the n th pulse of the alternating polarity pulses to reduce the net electrical charge delivered over the $n-1$ alternating polarity therapeutic pulses.
2. The medical device of claim 1, wherein the control circuit is further configured to determine a value of n based on an imbalanced charge of a pair of one positive polarity pulse and one negative polarity pulse of the alternating polarity pulses.
3. The medical device of claim 1, wherein the control circuit is further configured to control the therapy delivery circuit to modify the n th pulse of the alternating polarity pulses by reversing the polarity of the n th pulse.
4. The medical device of claim 1, wherein the control circuit is further configured to control the therapy delivery circuit to modify the n th pulse of the alternating polarity pulses by adjusting at least one of a pulse amplitude or a pulse width of the n th pulse.
5. The medical device of claim 1, further comprising a memory comprising a lookup table of pulse modifications, wherein the control circuit is further configured to: establish a modification of the n th pulse by retrieving the modification from the lookup table; and control the therapy delivery circuit to modify the n th pulse according to the established modification.
6. The medical device of claim 1, wherein: the therapy delivery circuit further comprises a first output circuit for generating electrical stimulation pulses and a second output circuit for generating electrical stimulation pulses; and the control circuit is further configured to: select one of the first output circuit and the second output circuit for generating the alternating polarity pulses, and establish at least one of a value of n or a modification of the n th pulse based on the selected one of the first output circuit and the second output circuit.
7. The medical device of claim 1, wherein the therapy delivery circuit is further configured to generate each of the alternating polarity pulses as biphasic pulses.
8. The medical device of claim 1, wherein the control circuit is further configured to: detect a change in an electrical stimulation output condition by detecting at least one of: a change in a cardiac pacing mode, a change in an electrode vector impedance, a change in a pulse amplitude of the electrical stimulation pulses, a change in a pulse width of the electrical stimulation pulses, or a change in a rate of the electrical stimulation pulses; and in response to detecting the change in the electrical stimulation output condition, change at least one of a value of n , a polarity of the n th pulse, a pulse amplitude of the n th pulse, or a pulse width of the n th pulse.
9. The medical device of claim 1, wherein the control circuit is further configured to: modify every n th pulse of the alternating polarity pulses according to a first modification and modify an m th pulse of the plurality of electrical stimulation pulses using a second modification different than the first modification.
10. The medical device of claim 1, wherein the therapy delivery circuit is configured to generate and deliver the electrical stimulation pulses as cardiac pacing pulses.
11. A non-transitory computer readable medium comprising instructions that, when executed by a

control circuit of a medical device, cause the medical device to: generate electrical stimulation pulses as alternating polarity pulses comprising $n-1$ alternating polarity therapeutic pulses that deliver a net electrical charge and an n th pulse, wherein the $n-1$ alternating polarity therapeutic pulses include at least one pair of opposite polarity therapeutic pulses such that n is at least 3; and modify the n th pulse of the alternating polarity pulses to reduce the net electrical charge delivered over the $n-1$ alternating polarity therapeutic pulses.

12. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to determine a value of n based on an imbalanced charge of a pair of one positive polarity pulse and one negative polarity pulse of the alternating polarity pulses.

13. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to modify the n th pulse of the alternating polarity pulses by reversing the polarity of the n th pulse.

14. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to modify the n th pulse of the alternating polarity pulses by adjusting at least one of a pulse amplitude or a pulse width of the n th pulse.

15. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to: establish a modification of the n th pulse by retrieving the modification from a lookup table; and modify the n th pulse according to the established modification.

16. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to: select one of a first output circuit and a second output circuit of the medical device for generating the alternating polarity pulses, and establish at least one of a value of n or a modification of the n th pulse based on the selected one of the first output circuit and the second output circuit.

17. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to generate each of the alternating polarity pulses as biphasic pulses.

18. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to: detect a change in an electrical stimulation output condition by detecting at least one of: a change in a cardiac pacing mode, a change in an electrode vector impedance, a change in a pulse amplitude of the electrical stimulation pulses, a change in a pulse width of the electrical stimulation pulses, or a change in a rate of the electrical stimulation pulses; and in response to detecting the change in the electrical stimulation output condition, change at least one of a value of n , a polarity of the n th pulse, a pulse amplitude of the n th pulse, or a pulse width of the n th pulse.

19. The non-transitory computer readable medium of claim 11 further comprising instructions that cause the medical device to: modify every n th pulse of the alternating polarity pulses according to a first modification and modify an m th pulse of the plurality of electrical stimulation pulses using a second modification different than the first modification.

20. A method comprising: generating electrical stimulation pulses as alternating polarity pulses by a therapy delivery circuit, the alternating polarity pulses comprising $n-1$ alternating polarity therapeutic pulses that deliver a net electrical charge and an n th pulse, wherein the $n-1$ alternating polarity therapeutic pulses include at least one pair of opposite polarity therapeutic pulses such that n is at least 3; and modifying the n th pulse of the alternating polarity pulses to reduce a net electrical charge delivered over the $n-1$ alternating polarity therapeutic pulses.
